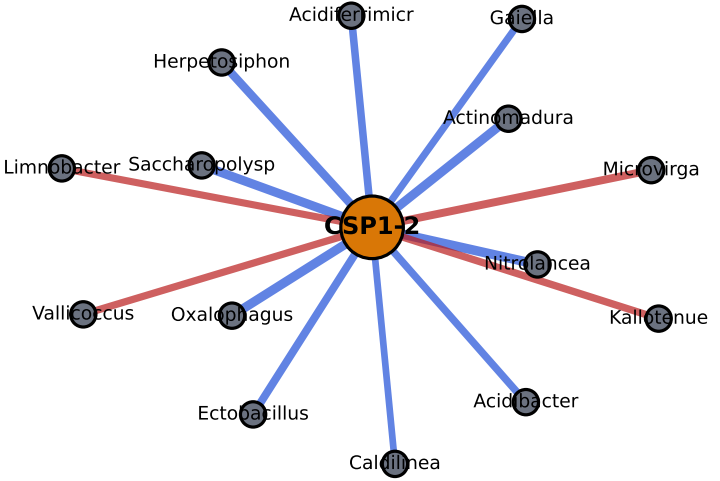
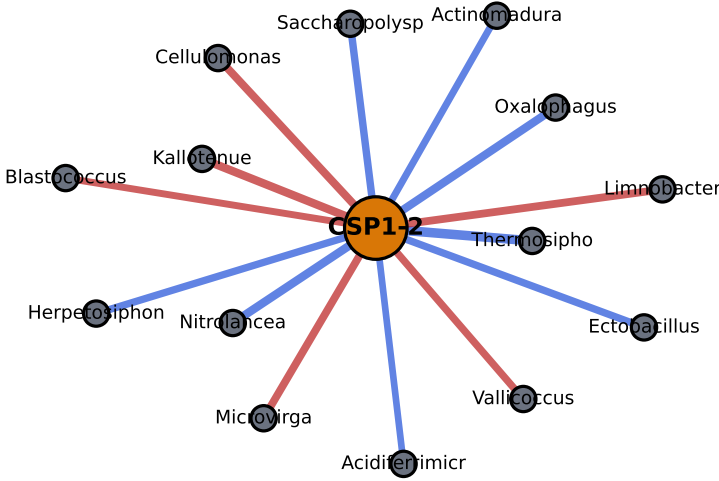


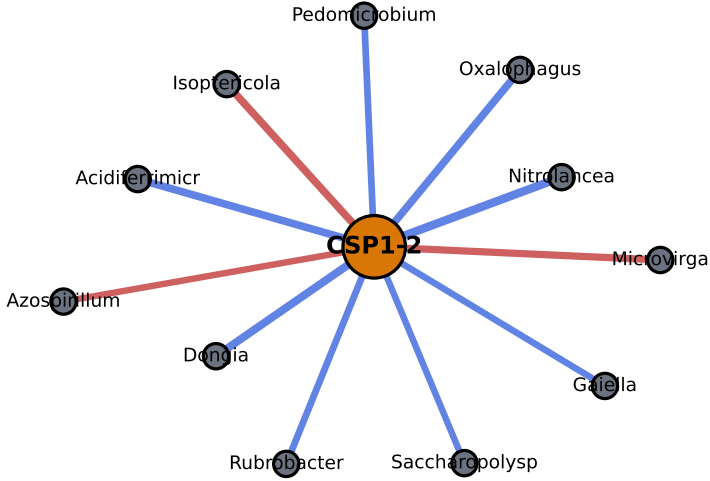
(a) Surface — CSP1-2 ego network (14 co-occurring genera)



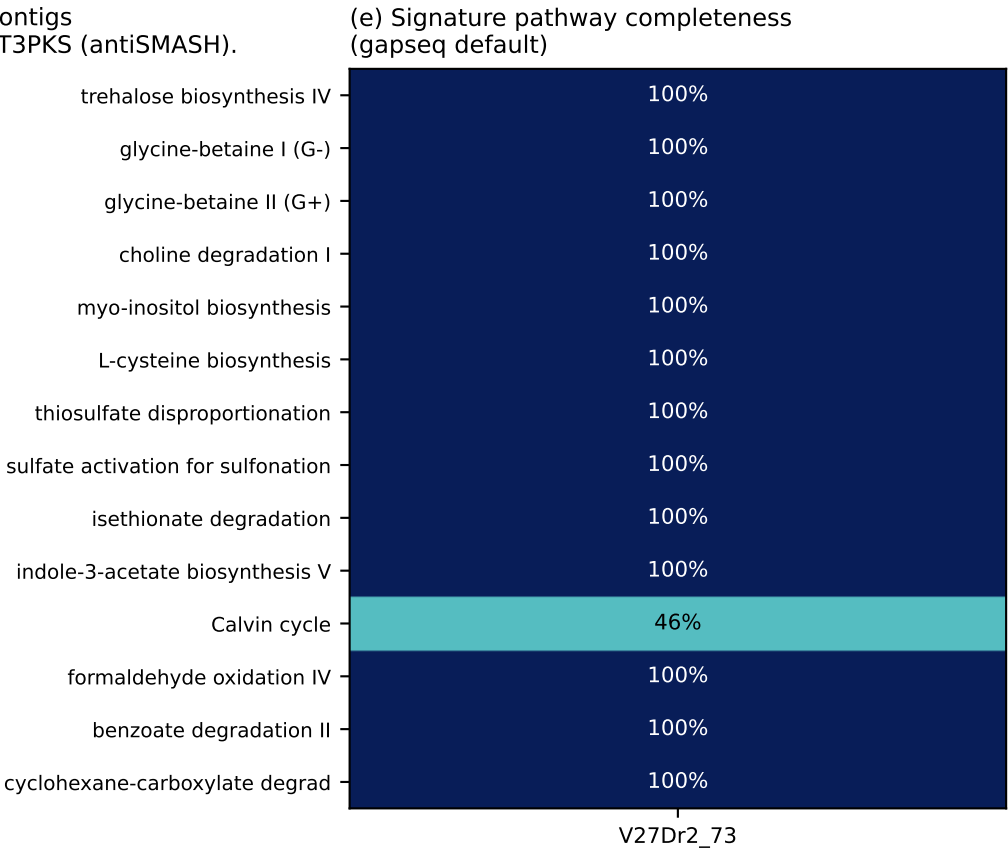
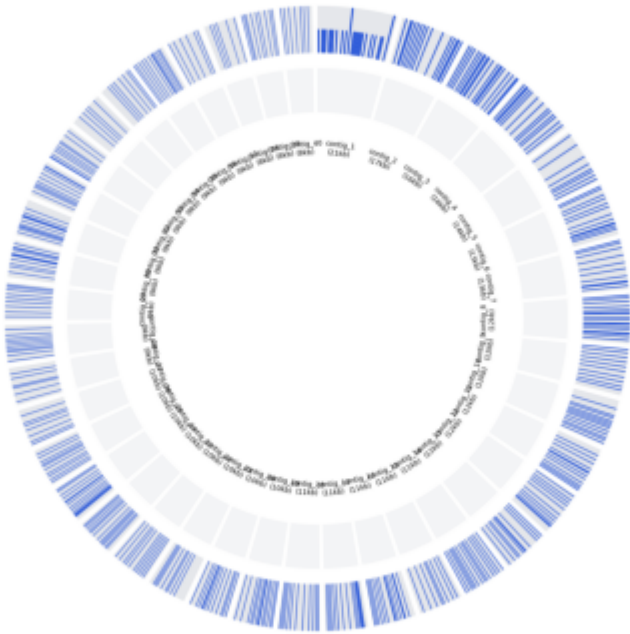
(b) Deep — CSP1-2 ego network (14 co-occurring genera)



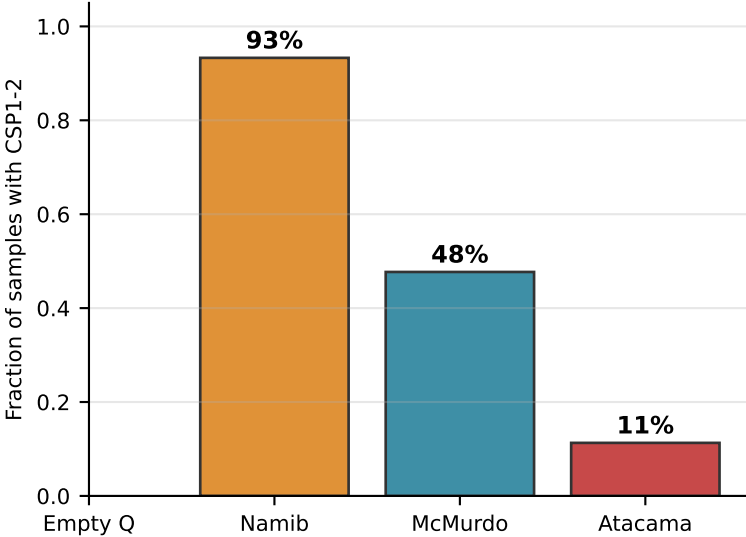
(c) Rhizosphere — CSP1-2 ego network (14 co-occurring genera)



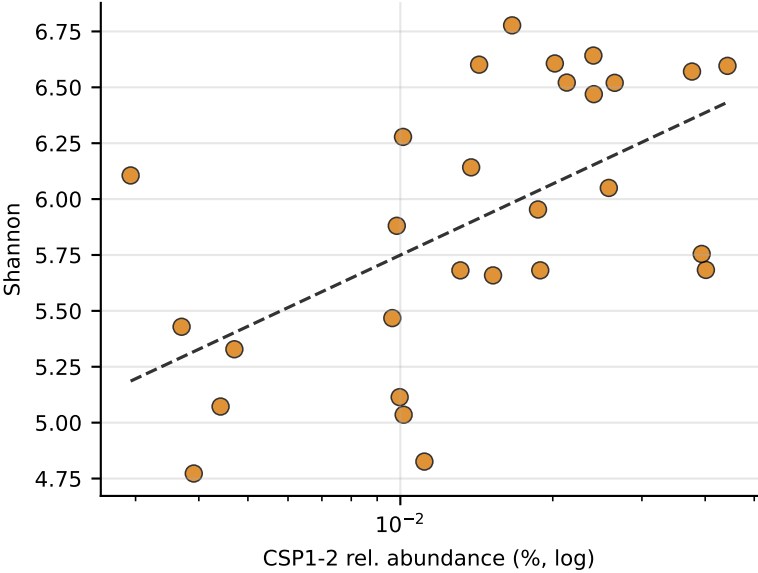
(d) Reference MAG V27Dr2_SemiBin_73 (2.66 Mb, GC 43.1%) — top 40 contigs
outer ring: CDS density (blue). 0 CRISPR arrays. BGCs: NRPS-like, RRE, T3PKS (antiSMASH).



(f) Cross-desert prevalence



(g) Namib — abundance tracks diversity
 $\rho = +0.64$, $p = 1.4 \times 10^{-4}$



(h) Atacama — salinity suppresses diversity
 $\rho = -0.38$, $p = 0.012$

